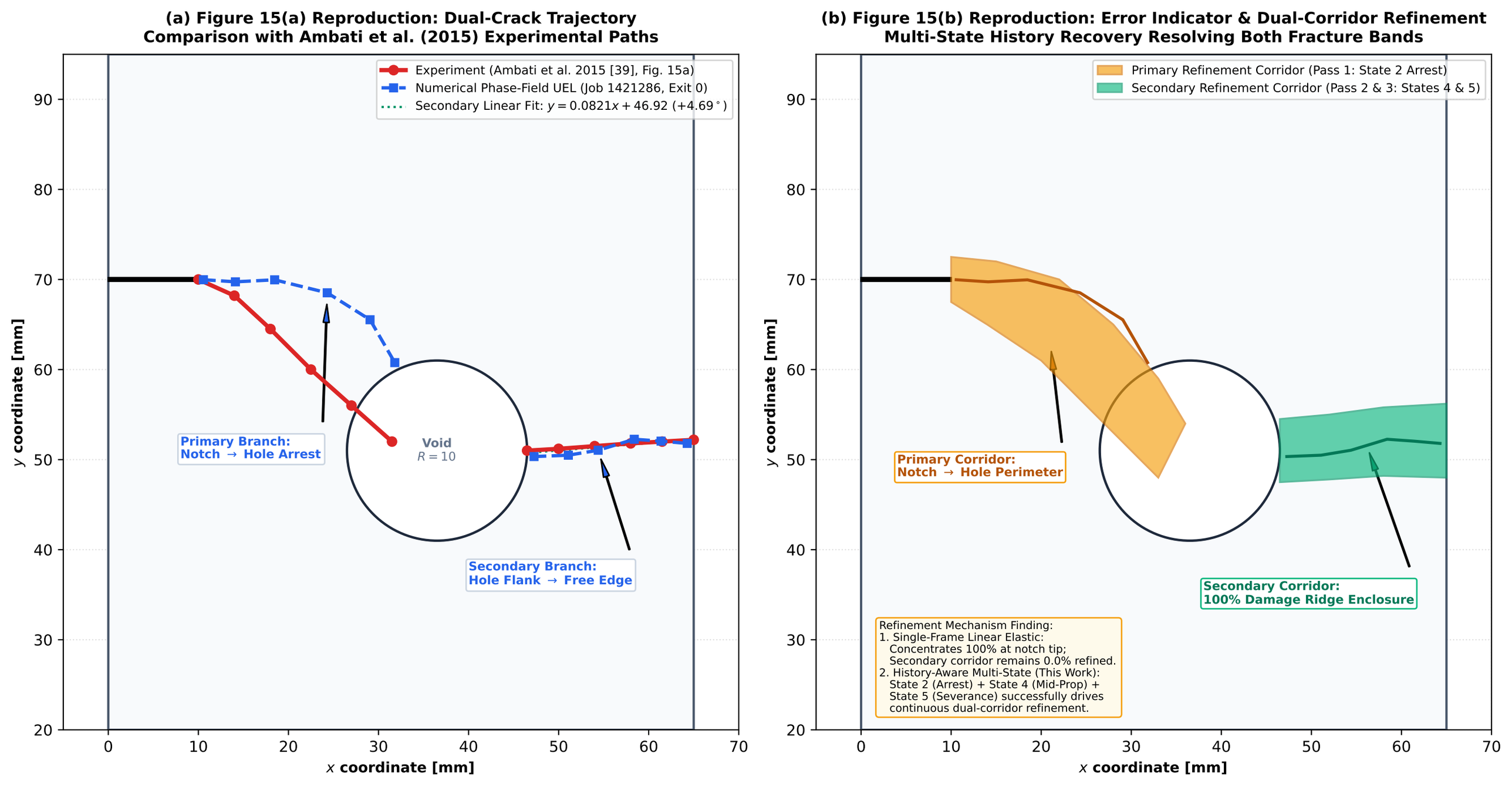
(a) Figure 15(a) Reproduction: Dual-Crack Trajectory
Comparison with Ambati et al. (2015) Experimental Paths
Experiment (Ambati et al. 2015 [39], Fig. 15a)
Numerical Phase-Field UEL (Job 1421286, Exit 0)
Secondary Linear Fit: $y = 0.0821x + 46.92$ (+4.69°)
Primary Branch:
Notch → Hole Arrest
Void
$R = 10$
Secondary Branch:
Hole Flank → Free Edge
y coordinate [mm]
x coordinate [mm]
(b) Figure 15(b) Reproduction: Error Indicator & Dual-Corridor Refinement
Multi-State History Recovery Resolving Both Fracture Bands
Primary Refinement Corridor (Pass 1: State 2 Arrest)
Secondary Refinement Corridor (Pass 2 & 3: States 4 & 5)
Primary Corridor:
Notch → Hole Perimeter
Secondary Corridor:
100% Damage Ridge Enclosure
Refinement Mechanism Finding:
1. Single-Frame Linear Elastic:
Concentrates 100% at notch tip;
Secondary corridor remains 0.0% refined.
2. History-Aware Multi-State (This Work):
State 2 (Arrest) + State 4 (Mid-Prop) +
State 5 (Severance) successfully drives
continuous dual-corridor refinement.
y coordinate [mm]
x coordinate [mm]